

A Literature Review on Low-Cost Smart Crutch for Safe Injury Rehabilitation and Recovery Monitoring

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Abstract— After a leg injury or operation, people need help with the weight they carry during walking. Because traditional crutches do not provide the feedback as to where you are applying pressure or how quickly you're prancing, following the rules of recovery can be challenging. It becomes guesswork. Without such data from things like those crutches, learning whether the movement is right turns uncertain. But the right amount of force to put in a step is rarely, particularly early recovery stages, and if used wrong, these devices can prolong healing and even cause fresh problems. In response to this issue, this research aims to create an smart crutch system that can detect the load and gait behavior of the user while delivering real time feedback through a vibration motor. Vibration feedback is incorporated in the prototype design as it can be felt with ease even in noisy or crowded or public environments. Using Bluetooth means data can be relayed to an app on a smartphone for graphical and progress tracking feedback showing the status of rehabilitation using a lightweight machine learning algorithm that assesses patient's metrics to predict their state of rehabilitation. The application contains a progress visualization, and reports can be sent to a doctor or care-giver. Wi-Fi (optional) makes it possible to monitor a person from the distance and gives access for several users on the system at the same time. The solution is designed for home and clinical rehabilitation with an opportunity of quality improvement of recovery while providing safety for a patient.

Keywords— *Rehabilitation, Smart Crutch, Weight Monitoring, Gait Analysis, Machine Learning, Mobile Application, Bluetooth Communication, Assistive Device.*

I. INTRODUCTION

Getting better after the broken bone or operation on the leg usually means doing rehab. At the starting, most folks lean on

the crutches to take stress off the hurt leg while staying mobile. Crutches are not always simple to manage. When directions are about unclear and someone might press down too hard or touch the supports and messing up the recovery speed. Attaching close to the advised load matters since going to the past it risks setbacks or fresh issues.

Mostly home rehab happens off the radar compared to the clinic visits. But when no one checks in regularly, tiny errors slip right through. Getting upright too fast post-injury? That split second counts. Wrong pressure can able to drag out recovery even with a spark new problem. Each small motion holds a secret if you pay attention. When actions are mindful instead of on repeat and healing gains ground. People using crutches at home often skip the extra support. Tools found in the clinics usually drain our pockets faster than common spending plans permit.

Mostly rehab systems today rely only on the sensors along with the smart devices to the support recovery. Yet a few of these tools are too much cost or confuse users. They struggle to blend into an everyday routine. When pressure sensors team up with the movement tracking and warnings pop up fast through a phone link. Alerts travel straight to the caregivers whether someone is resting at home or checked into care. Injuries such as broken bones or recovery after cuts into skin find aid here. This walking tool backs those mending from the lower limb harm. Built around steady progress and it fits the alongside rehab work without fuss.

II. LITERATURE REVIEW

Chalaki et al. [1] creates a smart walker system that used the camera data and fuzzy logic to assist the stroke patients. This system detect the movements and it enables one hand operation also. Main aim is to reduce the effort and improves the comfort while walking. Viegas et al. [2] looked into the setup for walkers that consist of load cells and optical sensors to monitor the walking. It detects the force distribution and

patterns while moving and enables the wireless monitoring to identify the improper usage of the setup.

Patel et al. [3] used the wearable sensors to monitor the patient recovery procedure in daily life and it tracks the body signals and moving patterns with outside of the clinical settings and it also supports the remote monitoring through the wireless data transmission procedure. Wei and Wu [4] also used the wearable sensors combined with the machine learning to see the rehabilitation analysis and it reviews the multiple sensor types and AI algorithms for the recovery monitoring process. It aims to predict the accurate diagnosis and with a personalized therapy through the data.

A study led by Guo et al. [5] introduced a shoe-based device is to track how the people walk, it helps doctors to judge about the healing progress in those dealing with long-term ankle issues. It apply the machine learning algorithm to analyse the movement and predict the recovery. This introduced system helps in the ankle rehabilitation progress. A device built by Naets et al. [6] the multiple sensors to track the patient movements and detect the walking errors and correct through providing the feedback during the rehabilitation and it improves the movement analysis. This device design is very cost effective also.

Megalingam, Greeshma, and Pillai's [7] gives the design of a smart crutch using the force sensors with accelerometers also. It work is to monitor weight and movement during the rehabilitation process and gives the instant alerts when the excessive pressure is applied on the smart crutch. From Qiu et al. [8], a tool called Gait Sense came with the wearable setup to monitor motion and pattern during the rehab. This tool used the multiple sensors to track the errors and gives the correct and low cost monitoring with outside of the clinical settings.

Similarly, motion sensors track leg movements during rehab sessions led by Monge and Postolache [9]. These devices also track the movement, heart rate and muscle activity also. It also supports the remote monitoring through the dashboard of online application and improves the user engagement. Girão and Postolache [10] explore the digital physiotherapy solution built on the S-Health platform using the IOT connected medical devices. This solution includes the smart walkers, crutches and sensors into the one platform and it also enables the remote monitoring and improves the rehabilitation through the continuous feedback.

Kyritsis et al. [11] built a way to recognize gait patterns using the smartphone motion sensors after the surgery or operation. Applied the machine learning technique to classify the different conditions of walking and also support the remote rehab monitoring without any additional hardware system. One study by Yang et al. [12] build a smart walking aid that helps the older adults or elderly people. It work is to track the locations, fall detection and monitors the signals and to be send the real time alerts based on the data and supports the remote supervision.

Postolache [13] developed one rehabilitation system using smart walkers and crutches with mobile apps and used the so

many sensors to see the walking patterns and also supports the real time data sharing to the apps to better tracking of the patient recovery. Chen et al. [14] designed smart crutches that capture 3D ground forces during when someone walks and used the machine learning (SVR) to calculate the weight distribution. It gives the vibration feedback and support the real world monitoring with low cost.

Lin et al. [15] created a tool called I Crutch - smart more than enough to connect to the smartphone mainly for the home assistance and used gesture based control to communicate with the smart home devices. This increases the safety and creates independence to the elderly people. Sarkar et al. [16], a smart crutch setup with IOT for the paralysis support and used sensors and GPS for any obstacle detection purpose and for the location tracking respectively. The design is a low-cost practical design with the remote monitoring feature.

Mukherjee et al. [17], a framework used beta regression to analyze the gait patterns. This work is based on the demographic and the movement factors and study about the combined effect of the parameters. It gives the accurate analysis of data of the gait patterns for rehabilitation. Sugiyama et al. [18] designed a crutch based training setup is to increase the balance control of the patient or user. It work is to improve the posture and recovery response when walking and stimulates in any falling conditions also.

Frango et al. [19] creates a smart crutch setup with IOT for rehabilitation monitoring and include the sensor to measure load and motion while the user walking. This work also supports the real time data transmission and remote monitoring through mobile app. Sardini et al. [20] build one smart crutch that send movement details wirelessly for a rehabilitation monitoring. That hardware design consist of a strain gauge and accelerometer and it gives the accurate data and the design is a low-cost solution for the clinical as well as home use.

Postolache [21] inserted force sensors into standard walker handles to monitor the load and balance of the user or patient and design consist with a electronic circuits to check the pressure changes during walking. This system helps to understand the movement patterns of the patient. Arcobelli et al. [22] creates "m-Crutch," based smart crutch mainly for remote monitoring and used motion and load sensors to track the tilt and pressure of the patient. This also supports the real time feedback and data sharing through the mobile app.

Resch et al. [23] combined the sensor insole and the smart crutch for rehabilitation support and gives the vibrational feedback to guide the patient regarding the weight distribution during walking. This design helps to increase the gait awareness through the real time feedback. Varghese et al. [24] purpose is to create a IOT based system to track the location with GPS and give the emergency alerts using the GSM. It gives the vibration feedback for safe weight.

Zhang and Liu's [25] introduced a smart cane design with consist of sensors for the safety and for the health monitoring. The design include with a Bluetooth to transmit the data to mobile app and to the remote server also. It also detects the

fall and obstacle detection while walking. Merrett et al. [26] designed a wireless gadget for forearm crutches to monitor the weight bearing and also used sensor to measure the pressure and tilt during recovery. It also gives the immediate feedback alerts for the correct usage of the user.

Payo et al. [27] created a simplified model to check the crutch assisted walking procedure. It divides the body and crutch into segments for the better estimation and gives the accurate output with simple setup. Jin's [28] studied about the rehabilitation for lower limb recovery after any stroke and focused on the accrete gait tracking data and motion analysis data. It is the combination of the biomechanics and robotics for increased therapy systems.

Tsuda et al. [29] study about the how much wrong crouches influences the body motion and aim to build a reliable assessment tool. It uses the tilt and motion data to detect the improper walking patterns of the patients while walking and also helps in increasing rehabilitation safety. Morris et al. [30] study about the challenges faced by the people with like impairments and highlight the limitations of existing devices in real life. The work emphasize the need for adaptable rehabilitation solutions.

Narvaez Dorado et al. [31] work also used the sensor-equipped crutches to check the different walking patterns. It measure the force, tilt and motion during recovery and helps in the accurate diagnosis and the rehabilitation monitoring. García-Barrios [32] explores how the AI supports the human decision making in tasks and highlight balance between the automation and the human control and submitted the importance of designing AI.

From the all reviews, almost existing systems are more complex and costly also. Based on these limitations, proposing a simple smart crutch with load and motion sensing with real-time feedback through vibration and app tracking also.

III. METHODOLOGY

The crutch monitors your every step with the aligned sensors. The weight the patient putting on the crutch will be detected by the load cell provided at the bottom of the crutch, that can firstly go through the HX711 amplifier because the load cell data is very weak. That data can't be processed by the heart of the system.

Then the data processed by the ESP32 module, with the Bluetooth connectivity the data transfer to the mobile application. In that application ML algorithm predicts the safe weight limit with the user personal information that he given before entering the dashboard. Based on the safe limit conditions the feedback is given through the vibration with another gait data also. Each data come to the application and it shows the recovery status in the graphical representation.

A. Load Measurement and Sensor Placement

A sensor built into the crutch system to check how much the person pushes down when stepping. Force turns into a small electric output, quietly showing exactly how strong the push was. When the signal comes through the amplifier, HX711 steps in - boosting strength while turning analog data into a controller understands to process. Because of this shift, tracking weight becomes very clear.

B. Processor and Data Handling

Inside the device, a small computer handles most of the work. From the HX711 sensor, it gives the number readings that show pressure. While showing these numbers, it watches closely if too much force appears during each step. Because of those checking's, output signals shift based on real-time balance needs. When danger occurs, vibration kicks in while information goes to your phone through Bluetooth. Real-time updates appear as alerts hints to guide the each step with precision to the patient. Feedback stays sharp because of the signals travel fast without any delay.

C. Feedback and Alert Mechanism

Vibration hints snap into place the moment things shift outside normal range. Because they depends on touch only, these pulses cut through street noise and busy scenes without needing eyes or ears. Walking adjustments happen faster when the signals get through skin instead of seeing or sound. Even in thick crowds, the body feels what the system saying.

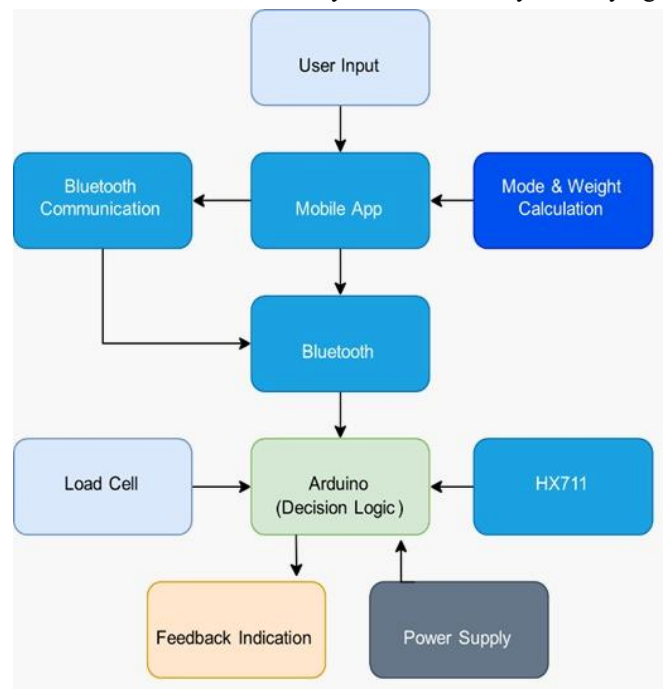


Figure 1: Block diagram of smart rehabilitation crutch system

D. Wireless Communication and Mobile Monitoring

Data moves from the crutch to a phone app through Bluetooth. As you move, measurements flow instantly into the software, showing the readings on screen. Past records stay saved inside the program. Recovery patterns become visible slowly, day by day. The device sends the data through wireless signals so recovery can be checked from home instead of going to clinics. Later on, if wanted, connecting via Wi-Fi could allow doctors to view updates and recovery.

E. Power Supply and Hardware Integration

In the device using a rechargeable battery for daily use and it provides the power to controller, sensors and other components in the hardware system. This maintain stable voltage to ensure perfect sensor readings. The device components are arranged compatly to reduce the power loss in the entire system.

IV. CONCLUSION

Recovery is not like just about the support, but it is about understanding each step taken while walking during the recovery. when the weight and movement are observed closely and even small improvements are also visible and meaningful. This work looks at how simple the sensing and the feedback can bring about the clarity to a process that is often based on the guesswork. Every step count when the smart crutch system measures pressure, movement and then replies in the right away. Instead of relying only on the daily periodic check-ups and users can able to receive guidance during everyday walking itself. The addition to the mobile tracking is allow progress to be seen by the overtime and turning the scattered efforts into a more connected recovery journey.

The system does not try to replace medical care, but slightly supports it by the filling the gaps between these sessions. Actually, this kind of assistance can help users to build their confidence, correct mistakes early, and move forward with some better control. Recovery becomes very clear when the technology responds in a real time.

V. FUTURE SCOPE

To improve the product, some features will be planned to integrate in future like integration of cloud storage for to store the data of users for a long term and it improves the multiple user access to the system. Development of doctor dashboard in the mobile application to doctor can access the the recovery Procedure of a patients. Accurate the ML algorithm prediction of the data in the mobile application.

Another one is to include the hospital ecosystem that contains the list of the hospitals. Under each hospital some patients will be there, that hospital system can access the all-patients recovery procedure. Improvement of mobile applicational features for better experience. Include the advanced sensors to get the advanced gait data, normal load cell data. Designing the hardware system with more user friendly and light weight and with a more compact design.

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